

Human Development Index (HDI) Predictor

Brainstorming & Idea Prioritization

Project: Human Development Index (HDI) Predictor

Phase: Brainstorming & Ideation

Developer: Nainaru Thulasi Ram

1. Ideas Considered

Idea	Domain	Feasibility	Impact
Human Development Index Predictor	Socio-economic analytics	High	High
Loan Approval Prediction	Finance	Medium	Medium
Hospital Management System	Healthcare ops	Medium	Medium
Flood Prediction System	Environmental	Low	High
Blood Donation Matching System	Healthcare/social	Medium	Low

2. Idea Prioritization Criteria

- Availability of a clear, measurable regression/classification target
- Strength of correlation between candidate input features and the target
- Feasibility of building a full pipeline (data -> model -> web app) within the internship timeline
- Educational value: coverage of EDA, preprocessing, model training, and deployment

3. Selected Idea

The Human Development Index (HDI) Predictor was selected as the highest-priority idea. It offered a well-defined statistical target (the HDI score), strongly correlated input indicators (life expectancy, education, income), and a natural end-to-end deliverable: a trained regression model exposed through a Flask web application.

4. Why Not the Alternatives

- Loan Approval Prediction — good fit for classification metrics, but less novel for a data-driven scoring demo
- Hospital Management System — more of a CRUD/database project than a machine learning project
- Flood Prediction System — requires time-series/geospatial data that was harder to source reliably
- Blood Donation Matching — smaller dataset scope, less rich feature set for regression